Class 12 Mathematics

Chapter 9: Differential Equations

Practice Set B: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE SQP 2026-27 — 1 mark

The degree and order of the differential equation $y - x\left(\frac{dy}{dx}\right)^3 = a\left(y + \frac{d^2y}{dx^2}\right)^2$ are respectively:

(A) $3,\ 1$ **(B)** $2,\ 2$ **(C)** $3,\ 2$ **(D)** $4,\ 2$

Formula / idea: Order = order of the highest derivative; degree = power of that derivative, once the equation is a polynomial in derivatives.

1 The highest order derivative is $\frac{d^2y}{dx^2}$.

$\text{Order} = 2$

2 The equation is a polynomial in derivatives. On expanding, $\frac{d^2y}{dx^2}$ has highest power 2. ✓ 1

$\text{Degree} = 2$

3 Degree first, then order: $2,\ 2$. Hence option (B).

Answer **(B)** $2,\ 2$

Watch out: Reading the degree as 3 from $\left(\frac{dy}{dx}\right)^3$; the degree is the power of the **highest order** derivative only.

2 MCQ NCERT Exemplar Q35 1 mark

The degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is:

(A) 4 **(B)** $\frac{3}{2}$ **(C)** not defined **(D)** 2

Formula / idea: Degree is a positive integer; first remove fractional powers of derivatives, then read the power of the highest derivative.

1 The power $\frac{3}{2}$ is not an integer. Square both sides to remove it.

$$\left[1+\left(\frac{dy}{dx}\right)^2\right]^3 = \left(\frac{d^2y}{dx^2}\right)^2$$

2 Now it is a polynomial in derivatives. The highest derivative $\frac{d^2y}{dx^2}$ has power 2. ✓ 1

$\text{Degree} = 2$

3 Hence option (D).

Answer **(D)** 2

Watch out: Answering $\frac{3}{2}$ or 'not defined'; a fractional power is removed by squaring, and the degree is then 2.

3 MCQ NCERT Exemplar Q55 1 mark

The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is:

(A) $\frac{x}{e^x}$ **(B)** $\frac{e^x}{x}$ **(C)** xe^x **(D)** e^x

Formula / idea: For $\frac{dy}{dx} + Py = Q$, $\text{I.F.} = e^{\int P\,dx}$.

1 Split the right side and bring the y term to the left.

$$\frac{dy}{dx} + \left(1 - \frac{1}{x}\right)y = \frac{1}{x}$$

2 Compare with $\frac{dy}{dx} + Py = Q$ and integrate P.

$P = 1 - \frac{1}{x}$, $\int P\,dx = x - \log x$

3 Find the integrating factor. ✓ 1

$\text{I.F.} = e^{x-\log x} = e^x \cdot e^{-\log x} = \frac{e^x}{x}$

4 Hence option (B).

Answer (B) $\dfrac{e^x}{x}$

Watch out: Taking $P = 1$ and choosing e^x; the $\dfrac{y}{x}$ hidden in $\dfrac{1+y}{x}$ is also part of P.

4 MCQ Board-pattern 1 mark

The number of arbitrary constants in the general solution of the differential equation $\left(\dfrac{d^2y}{dx^2}\right)^3 + \left(\dfrac{dy}{dx}\right)^4 + y = e^x$ is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Formula / idea: The general solution has as many arbitrary constants as the order of the equation.

1 The highest order derivative is $\dfrac{d^2y}{dx^2}$.
$\text{Order} = 2$

2 Number of arbitrary constants = order $= 2$. ✓ 1

3 Hence option (B).

Answer (B) 2

Watch out: Using the degree 3 or the power 4; the number of constants depends only on the order.

5 MCQ NCERT Exemplar Q73 1 mark

The general solution of the differential equation $(e^x + 1)\, y\, dy = (y+1)\, e^x\, dx$ is:

(A) $(y+1) = k\,(e^x + 1)$ **(B)** $y + 1 = e^x + 1 + k$

(C) $y = \log\{k(y+1)\,(e^x+1)\}$ **(D)** $y = \log\left\{\dfrac{e^x+1}{y+1}\right\} + k$

Formula / idea: Variable separable: put all y terms with dy and all x terms with dx, then integrate.

1 Separate the variables.
$\dfrac{y}{y+1}\,dy = \dfrac{e^x}{e^x+1}\,dx$

2 Write $\dfrac{y}{y+1} = 1 - \dfrac{1}{y+1}$ and integrate both sides. ✓ 1
$y - \log|y+1| = \log(e^x+1) + \log k$

3 Move $\log|y+1|$ to the right and combine the logs.
$y = \log\{k(y+1)\,(e^x+1)\}$

4 Hence option (C).

Answer **(C)** $y = \log\{k(y+1)(e^x+1)\}$

Watch out: Moving $-\log|y+1|$ across without changing its sign, which gives option (D).

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): $y = ae^x + be^x$, where a and b are arbitrary constants, is the general solution of the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$. **Reason (R):** The general solution of a differential equation contains as many arbitrary constants as the order of the equation.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) **Assertion (A) is false, but Reason (R) is true.**

1 Combine the terms: $a + b$ is one constant A.
$y = (a+b)e^x = Ae^x$

2 Check: $y' = y'' = Ae^x$, so y satisfies the equation.
$Ae^x - 2Ae^x + Ae^x = 0$

3 It has only one arbitrary constant, but the order is 2. So A is false.

4 R is the definition of a general solution. So R is true.

5 A is false and R is true. Hence option (D). ✓ 1

Answer **(D) A is false, but R is true.**

Watch out: Counting a and b as two constants; $ae^x + be^x = (a+b)e^x$ has only one.

7 **Short answer** NCERT Misc. Ex. Q4 2 marks

Find the general solution of the differential equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$.

Formula / idea: $\int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + C$

1 Separate the variables.
$\frac{dy}{\sqrt{1-y^2}} = -\frac{dx}{\sqrt{1-x^2}}$ ✓ ½

2 Integrate both sides. ✓ 1

$$\int \frac{dy}{\sqrt{1-y^2}} = -\int \frac{dx}{\sqrt{1-x^2}} \Rightarrow \sin^{-1} y = -\sin^{-1} x + C$$

3 Hence the general solution is $\sin^{-1} x + \sin^{-1} y = C$. ✓ ½

Answer $\sin^{-1} x + \sin^{-1} y = C$

Watch out: Losing the minus sign when moving $\sqrt{\frac{1-y^2}{1-x^2}}$ to the right side.

Level 2

18 marks

8 **Short answer** Board-pattern 2 marks

The population of a town increases at a rate proportional to the population present at that time, the constant of proportionality being k $(k > 0)$. Show that the population doubles in time $\frac{\log 2}{k}$, whatever the initial population.

Formula / idea: Rate proportional to amount: $\frac{dP}{dt} = kP$ gives $P = P_0 e^{kt}$.

1 Let P be the population at time t, and P_0 at $t = 0$. ✓ ½

$$\frac{dP}{dt} = kP$$

2 Separate and integrate.

$$\frac{dP}{P} = k\,dt \Rightarrow \log P = kt + C$$

3 At $t = 0$, $P = P_0$, so $C = \log P_0$. ✓ ½

$$\log \frac{P}{P_0} = kt \Rightarrow P = P_0 e^{kt}$$

4 Let T be the doubling time. Put $P = 2P_0$. ✓ ½

$$2P_0 = P_0 e^{kT} \Rightarrow e^{kT} = 2$$

5 Take log. P_0 has cancelled, so T does not depend on the initial population. ✓ ½

$$T = \frac{\log 2}{k}$$

Answer $P = P_0 e^{kt}$; **doubling time** $T = \frac{\log 2}{k}$, **which does not involve** P_0

Watch out: Putting $P = 2$ instead of $P = 2P_0$; doubling means twice the starting value.

9 **Short answer** NCERT Misc. Ex. Q12 3 marks

Find a particular solution of the differential equation $(x+1)\dfrac{dy}{dx} = 2e^{-y} - 1$, given that $y = 0$ when $x = 0$.

Formula / idea: $\displaystyle\int \frac{f'(y)}{f(y)}\,dy = \log|f(y)| + C$

1	Write the right side as one fraction. $(x+1)\dfrac{dy}{dx} = \dfrac{2 - e^y}{e^y}$	
2	Separate the variables. $\dfrac{e^y}{2 - e^y}\,dy = \dfrac{dx}{x+1}$	✓ ½
3	For the left side, put $2 - e^y = t$, so $-e^y\,dy = dt$. $\displaystyle\int \frac{e^y}{2-e^y}\,dy = -\int \frac{dt}{t} = -\log|2 - e^y|$	✓ ½
4	Integrate both sides. $-\log|2 - e^y| = \log|x+1| + C$	✓ ½
5	Put $x = 0, y = 0$. $-\log 1 = \log 1 + C \;\Rightarrow\; C = 0$	✓ ½
6	So $\log|2 - e^y| + \log|x+1| = 0$. Near $(0,0)$ both factors are positive. $(2 - e^y)(x+1) = 1 \;\Rightarrow\; e^y = 2 - \dfrac{1}{x+1} = \dfrac{2x+1}{x+1}$	✓ ½
7	Take log. Hence the particular solution is $y = \log\left\|\dfrac{2x+1}{x+1}\right\|$. $y = \log\left\|\dfrac{2x+1}{x+1}\right\|$	✓ ½

Answer $y = \log\left|\dfrac{2x+1}{x+1}\right|$

Watch out: Writing $\displaystyle\int \frac{e^y}{2-e^y}\,dy = \log|2 - e^y|$; the derivative of $2 - e^y$ is $-e^y$, so a minus sign appears.

10 **Short answer** NCERT Misc. Ex. Q10 3 marks

Solve the differential equation $\left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right]\frac{dx}{dy} = 1 \ \ (x \neq 0)$.

Formula / idea: For $\frac{dy}{dx} + Py = Q$: $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\, dx + C$, where $\text{I.F.} = e^{\int P\, dx}$.

1 Take the reciprocal to get $\frac{dy}{dx}$. ✓ ½

$$\frac{dy}{dx} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}$$

2 Rearrange: it is linear in y. ✓ ½

$$\frac{dy}{dx} + \frac{1}{\sqrt{x}}y = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}, \qquad P = \frac{1}{\sqrt{x}}$$

3 Find the integrating factor. ✓ ½

$$\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} \ \Rightarrow \ \text{I.F.} = e^{2\sqrt{x}}$$

4 Write the solution formula. ✓ ½

$$y\, e^{2\sqrt{x}} = \int \frac{e^{-2\sqrt{x}}}{\sqrt{x}} \cdot e^{2\sqrt{x}}\, dx + C$$

5 The exponentials cancel. Integrate. ✓ ½

$$y\, e^{2\sqrt{x}} = \int x^{-\frac{1}{2}}\, dx = 2\sqrt{x} + C$$

6 Hence the general solution is $y\, e^{2\sqrt{x}} = 2\sqrt{x} + C$. ✓ ½

Answer $y\, e^{2\sqrt{x}} = 2\sqrt{x} + C$

Watch out: Treating it as linear in x because $\frac{dx}{dy}$ is given; take the reciprocal first.

11 **Short answer** NCERT Exemplar Q16 3 marks

Solve the differential equation $x^2\frac{dy}{dx} = x^2 + xy + y^2$.

Formula / idea: Homogeneous equation $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$: put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

1 Divide by x^2. The right side is a function of $\frac{y}{x}$, so it is homogeneous. ✓ ½

$$\frac{dy}{dx} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2$$

2 Put $y = vx$. ✓ ½

$$v + x\frac{dv}{dx} = 1 + v + v^2$$

3	Cancel v on both sides. $x\frac{dv}{dx} = 1 + v^2$	✓ ½
4	Separate the variables. $\frac{dv}{1+v^2} = \frac{dx}{x}$	✓ ½
5	Integrate both sides. $\tan^{-1} v = \log\lvert x\rvert + C$	✓ ½
6	Put $v = \frac{y}{x}$. Hence the general solution is $\tan^{-1}\left(\frac{y}{x}\right) = \log\lvert x\rvert + C$.	✓ ½

Answer $\tan^{-1}\left(\frac{y}{x}\right) = \log\lvert x\rvert + C$

Watch out: Forgetting to cancel v from both sides and getting $x\frac{dv}{dx} = 1 + v + v^2$.

12 **Short answer** NCERT Misc. Ex. Q2(i) 2 marks

Verify that the function $xy = ae^x + be^{-x} + x^2$ is a solution of the differential equation $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0$.

Formula / idea: A function is a solution if it satisfies the equation identically.

1	Differentiate both sides w.r.t. x (product rule on the left). $x\frac{dy}{dx} + y = ae^x - be^{-x} + 2x$	✓ ½
2	Differentiate again w.r.t. x. $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = ae^x + be^{-x} + 2$	✓ ½
3	From the given function, $ae^x + be^{-x} = xy - x^2$. Substitute. $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = xy - x^2 + 2$	✓ ½
4	Rearrange. Hence verified. $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0$	✓ ½

Answer $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0$ **holds, so the function is a solution.**

Watch out: Writing $\frac{d}{dx}(xy) = \frac{dy}{dx}$; by the product rule it is $x\frac{dy}{dx} + y$.

13 **Short answer** NCERT Ex 9.3 Q1 2 marks

Find the general solution of the differential equation $\dfrac{dy}{dx} = \dfrac{1-\cos x}{1+\cos x}$.

Formula / idea: $1-\cos x = 2\sin^2\dfrac{x}{2}$, $1+\cos x = 2\cos^2\dfrac{x}{2}$ and $\tan^2\theta = \sec^2\theta - 1$.

1 Use the half-angle formulas. ✓ ½

$$\frac{1-\cos x}{1+\cos x} = \frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}} = \tan^2\frac{x}{2}$$

2 Write $\tan^2\dfrac{x}{2} = \sec^2\dfrac{x}{2} - 1$ and separate. ✓ ½

$$dy = \left(\sec^2\frac{x}{2} - 1\right)dx$$

3 Integrate both sides. ✓ ½

$$y = 2\tan\frac{x}{2} - x + C$$

4 Hence the general solution is $y = 2\tan\dfrac{x}{2} - x + C$. ✓ ½

Answer $y = 2\tan\dfrac{x}{2} - x + C$

Watch out: Writing $\int \sec^2\dfrac{x}{2}\,dx = \tan\dfrac{x}{2}$; dividing by the coefficient $\dfrac{1}{2}$ gives $2\tan\dfrac{x}{2}$.

14 **Short answer** NCERT Exemplar Q12 3 marks

If $y(t)$ is a solution of $(1+t)\dfrac{dy}{dt} - ty = 1$ and $y(0) = -1$, then show that $y(1) = -\dfrac{1}{2}$.

Formula / idea: For $\dfrac{dy}{dt} + Py = Q$: $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dt + C$, where $\text{I.F.} = e^{\int P\,dt}$.

1 Divide by $1+t$ to get the standard form. ✓ ½

$$\frac{dy}{dt} - \frac{t}{1+t}y = \frac{1}{1+t}$$

2 Here $P = -\dfrac{t}{1+t} = -1 + \dfrac{1}{1+t}$. Integrate P. ✓ ½

$$\int P\,dt = -t + \log(1+t)$$

$t = (1+t) - 1$, so $\dfrac{t}{1+t} = 1 - \dfrac{1}{1+t}$

3 Find the integrating factor. ✓ ½

$$\text{I.F.} = e^{-t+\log(1+t)} = (1+t)\,e^{-t}$$

4 Write the solution formula and integrate. ✓ ½

$$y(1+t)e^{-t} = \int \frac{1}{1+t}\cdot(1+t)e^{-t}\,dt = -e^{-t} + C$$

5 Put $t = 0, y = -1$.
$-1 \cdot 1 \cdot 1 = -1 + C \Rightarrow C = 0$ ✓ ½

6 So $y = -\dfrac{1}{1+t}$. Put $t = 1$. Hence shown.
$y(1) = -\dfrac{1}{1+1} = -\dfrac{1}{2}$ ✓ ½

Answer $y = -\dfrac{1}{1+t}$, **so** $y(1) = -\dfrac{1}{2}$

Watch out: Taking $P = -t$ without dividing the whole equation by $(1+t)$ first.

Level 3

25 marks

15 **Long answer** NCERT Ex 9.4 Q9 5 marks

Show that the differential equation $y\,dx + x\log\left(\dfrac{y}{x}\right)dy - 2x\,dy = 0$ is homogeneous and solve it.

Formula / idea: Homogeneous: $F(\lambda x, \lambda y) = \lambda^0 F(x, y)$. Put $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.

1 Write $\dfrac{dy}{dx} = F(x, y)$.
$\dfrac{dy}{dx} = \dfrac{y}{2x - x\log\left(\frac{y}{x}\right)} = F(x, y)$ ✓ ½

2 Replace x, y by $\lambda x, \lambda y$; $\dfrac{y}{x}$ does not change. So it is homogeneous.
$F(\lambda x, \lambda y) = \dfrac{\lambda y}{\lambda x\left(2 - \log\frac{y}{x}\right)} = \lambda^0 F(x, y)$ ✓ ½

3 Put $y = vx$.
$v + x\dfrac{dv}{dx} = \dfrac{v}{2 - \log v}$ ✓ ½

4 Move v to the right.
$x\dfrac{dv}{dx} = \dfrac{v - 2v + v\log v}{2 - \log v} = \dfrac{v(\log v - 1)}{2 - \log v}$ ✓ ½

5 Separate the variables.
$\dfrac{2 - \log v}{v(\log v - 1)}\,dv = \dfrac{dx}{x}$ ✓ ½

6 Put $\log v - 1 = t$, so $\dfrac{dv}{v} = dt$ and $2 - \log v = 1 - t$.
$\displaystyle\int \frac{1-t}{t}\,dt = \int \frac{dx}{x}$ ✓ ½

7 Integrate both sides.
$\log|t| - t = \log|x| + C_1$ ✓ 1

8 Collect the logs and take exponentials. ✓ ½

$$\log\left|\frac{t}{x}\right| = t + C_1 \;\Rightarrow\; \frac{t}{x} = A\,e^{t}$$

$A = \pm e^{C_1}$

9 Here $e^{t} = e^{\log v - 1} = \dfrac{v}{e} = \dfrac{y}{ex}$. Substitute.

$$\frac{t}{x} = \frac{A\,y}{e\,x} \;\Rightarrow\; t = \frac{A}{e}\,y$$

10 Put $t = \log\dfrac{y}{x} - 1$ and $c = \dfrac{A}{e}$. Hence the general solution is $cy = \log\left|\dfrac{y}{x}\right| - 1$. ✓ ½

$$cy = \log\left|\frac{y}{x}\right| - 1$$

Answer $cy = \log\left|\dfrac{y}{x}\right| - 1$

Watch out: Splitting $\dfrac{1-t}{t}$ wrongly; it is $\dfrac{1}{t} - 1$, which integrates to $\log|t| - t$.

16 Short answer NCERT Misc. Ex. Q11 — 3 marks

Find a particular solution of the differential equation $\dfrac{dy}{dx} + y\cot x = 4x\,\operatorname{cosec} x \;\; (x \neq 0)$, given that $y = 0$ when $x = \dfrac{\pi}{2}$.

Formula / idea: For $\dfrac{dy}{dx} + Py = Q$: $y \cdot \text{I.F.} = \displaystyle\int Q \cdot \text{I.F.}\, dx + C$, where $\text{I.F.} = e^{\int P\,dx}$.

1 Compare with $\dfrac{dy}{dx} + Py = Q$. ✓ ½

$$P = \cot x, \quad Q = 4x\,\operatorname{cosec} x$$

2 Find the integrating factor. ✓ ½

$$\text{I.F.} = e^{\int \cot x\,dx} = e^{\log \sin x} = \sin x$$

3 Write the solution formula; $\operatorname{cosec} x \cdot \sin x = 1$. ✓ ½

$$y\sin x = \int 4x\,\operatorname{cosec} x \cdot \sin x\,dx = \int 4x\,dx$$

4 Integrate. ✓ ½

$$y\sin x = 2x^2 + C$$

5 Put $x = \dfrac{\pi}{2}$, $y = 0$. ✓ ½

$$0 = 2\cdot\frac{\pi^2}{4} + C \;\Rightarrow\; C = -\frac{\pi^2}{2}$$

6 Hence the particular solution is $y \sin x = 2x^2 - \dfrac{\pi^2}{2}$. ✓ ½

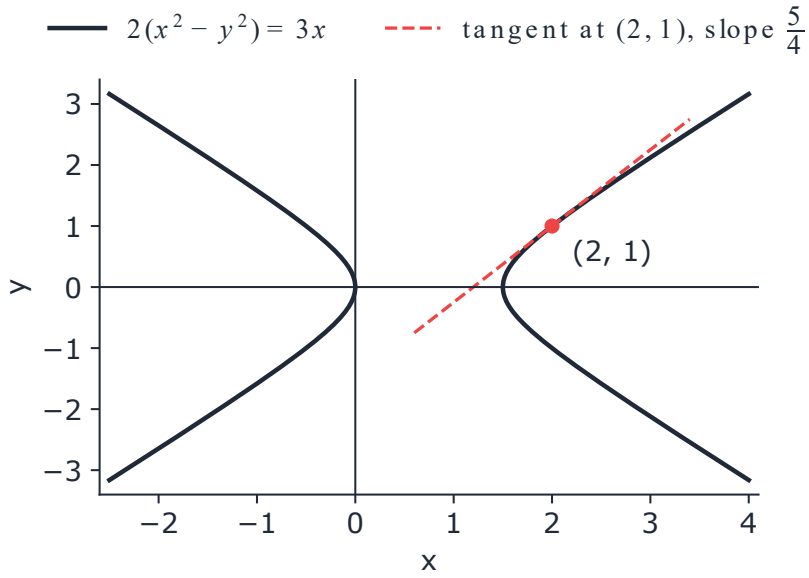
Answer $y \sin x = 2x^2 - \dfrac{\pi^2}{2}$

Watch out: Writing $\text{I.F.} = e^{\int \cot x\, dx} = \cos x$; the integral of $\cot x$ is $\log \sin x$, so $\text{I.F.} = \sin x$.

17 **Long answer** NCERT Exemplar Q29 5 marks

Find the equation of a curve passing through the point $(2, 1)$ if the slope of the tangent to the curve at any point (x, y) is $\dfrac{x^2 + y^2}{2xy}$.

Formula / idea: Slope of the tangent $= \dfrac{dy}{dx}$. A homogeneous equation is solved by $y = vx$.



1 Slope of the tangent is $\dfrac{dy}{dx}$.
$\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{2xy} = F(x, y)$ ✓ ½

2 Replace x, y by $\lambda x, \lambda y$. So it is homogeneous.
$F(\lambda x, \lambda y) = \dfrac{\lambda^2 (x^2 + y^2)}{\lambda^2 \cdot 2xy} = \lambda^0 F(x, y)$ ✓ ½

3 Put $y = vx$.
$v + x\dfrac{dv}{dx} = \dfrac{1 + v^2}{2v}$ ✓ ½

4 Move v to the right.
$x\dfrac{dv}{dx} = \dfrac{1 + v^2 - 2v^2}{2v} = \dfrac{1 - v^2}{2v}$ ✓ ½

5 Separate the variables. ✓ ½
$$\frac{2v}{1-v^2}\,dv = \frac{dx}{x}$$

6 Integrate both sides. ✓ 1
$$-\log|1-v^2| = \log|x| + C_1$$
Put $1 - v^2 = t$, so $-2v\,dv = dt$.

7 Combine the logs.
$$x\left(1-v^2\right) = C$$
$$C = \pm e^{-C_1}$$

8 Put $v = \frac{y}{x}$. ✓ ½
$$x\left(1-\frac{y^2}{x^2}\right) = C \;\Rightarrow\; x^2 - y^2 = Cx$$

9 The curve passes through $(2, 1)$. ✓ ½
$$4 - 1 = 2C \;\Rightarrow\; C = \frac{3}{2}$$

10 Hence the required curve is $2(x^2 - y^2) = 3x$. ✓ ½
$$2\left(x^2 - y^2\right) = 3x$$

Answer $2(x^2 - y^2) = 3x$

Watch out: Writing $\int \frac{2v}{1-v^2}\,dv = \log|1-v^2|$; the derivative of $1 - v^2$ is $-2v$, so it is $-\log|1-v^2|$.

18 **Short answer** CBSE SQP 2026-27 3 marks

Find the particular solution of the differential equation $(x^2 + y^2)\,dy = xy\,dx$, given that $y(0) = 1$.

Formula / idea: Homogeneous equation: put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

1 Write $\frac{dy}{dx}$. Both parts have degree 2, so it is homogeneous. ✓ ½
$$\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$$

2 Put $y = vx$.
$$v + x\frac{dv}{dx} = \frac{v}{1+v^2}$$

3 Move v to the right. ✓ ½
$$x\frac{dv}{dx} = \frac{v - v - v^3}{1+v^2} = -\frac{v^3}{1+v^2}$$

4 Separate the variables.
$$\left(\frac{1}{v^3} + \frac{1}{v}\right)dv = -\frac{dx}{x}$$

5 Integrate both sides. ✓ ½

$$-\frac{1}{2v^2} + \log|v| = -\log|x| + C$$

6 Put $v = \frac{y}{x}$; $\log|x|$ cancels. ✓ ½

$$-\frac{x^2}{2y^2} + \log|y| - \log|x| = -\log|x| + C \;\Rightarrow\; \log|y| - \frac{x^2}{2y^2} = C$$

7 Put $x = 0, y = 1$. ✓ ½

$$\log 1 - 0 = C \;\Rightarrow\; C = 0$$

8 Hence the particular solution is $x^2 = 2y^2 \log|y|$. ✓ ½

$$\log|y| = \frac{x^2}{2y^2} \;\Rightarrow\; x^2 = 2y^2 \log|y|$$

Answer $x^2 = 2y^2 \log|y|$, **i.e.** $y = e^{\frac{x^2}{2y^2}}$

Watch out: Writing $\int \frac{dv}{v^3} = \frac{1}{2v^2}$; it is $-\frac{1}{2v^2}$.

19 **Long answer** CBSE SQP 2025-26 5 marks

Solve the differential equation $y + \frac{d}{dx}(xy) = x(\sin x + x)$.

Formula / idea: For $\frac{dy}{dx} + Py = Q$: $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\, dx + C$, where $\text{I.F.} = e^{\int P\, dx}$.

1 Expand $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$. ✓ ½

$$y + x\frac{dy}{dx} + y = x(\sin x + x) \;\Rightarrow\; x\frac{dy}{dx} + 2y = x(\sin x + x)$$

2 Divide by x: it is linear in y. ✓ ½

$$\frac{dy}{dx} + \frac{2}{x}y = \sin x + x$$

3 Here $P = \frac{2}{x}$. Find the integrating factor. ✓ 1

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2\log x} = x^2$$

4 Write the solution formula. ✓ 1

$$y \cdot x^2 = \int x^2(\sin x + x)\, dx = \int x^2 \sin x\, dx + \int x^3\, dx$$

5 Integrate $x^2 \sin x$ by parts. ✓ ½

$$\int x^2 \sin x\, dx = -x^2 \cos x + \int 2x \cos x\, dx$$

6 Integrate $2x \cos x$ by parts again. ✓ ½

$$\int 2x \cos x\, dx = 2\,(x \sin x + \cos x)$$

7 Add $\int x^3\,dx = \dfrac{x^4}{4}$. Hence the general solution is

$$x^2y = -x^2\cos x + 2x\sin x + 2\cos x + \frac{x^4}{4} + C$$

✓ 1

Answer $x^2y = -x^2\cos x + 2x\sin x + 2\cos x + \dfrac{x^4}{4} + C$

Watch out: Writing $\dfrac{d}{dx}(xy) = x\dfrac{dy}{dx}$ and missing one y; then $P = \dfrac{1}{x}$ and the I.F. is wrong.

20 **Case study** Board-pattern 4 marks

Case Study: After a medicine is injected, the body removes it from the blood at a rate proportional to the amount present at that time. A patient is given a dose of 400 mg. After 3 hours, 300 mg of the medicine is still present. Let A mg be the amount present t hours after the dose.

(i) Write the differential equation satisfied by A and state its order and degree. 1 mark

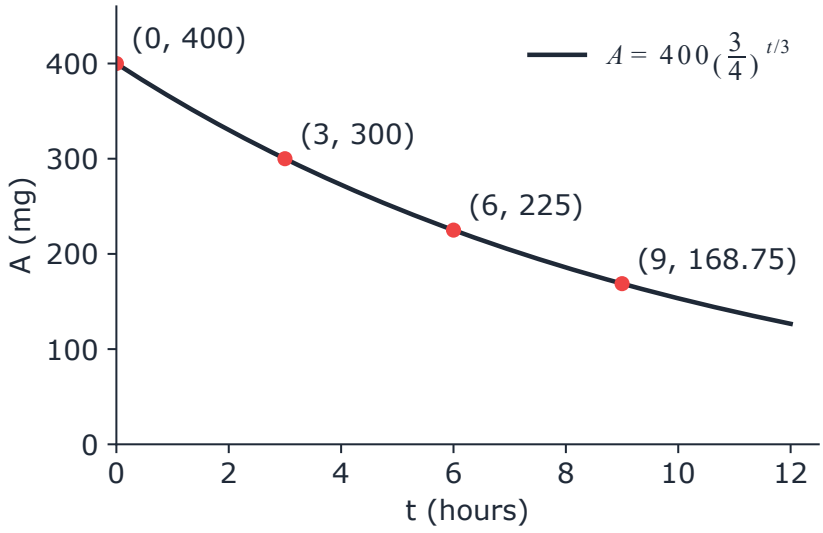
(ii) Show that $A = 400\left(\dfrac{3}{4}\right)^{t/3}$. 1 mark

(iii)(a) Find the amount of medicine present 6 hours after the dose, and the percentage of the dose removed by then. 2 marks

OR

(iii)(b) The next dose is given when the amount falls to 168.75 mg. After how many hours is the next dose given? 2 marks

Formula / idea: Rate proportional to amount: $\dfrac{dA}{dt} = -kA$ gives $A = A_0e^{-kt}$.



(i) 1 mark

1 The amount decreases, so the rate is negative. Take constant $k > 0$.

$$\frac{dA}{dt} = -kA$$

✓ ½

2 Highest derivative is $\frac{dA}{dt}$, with power 1.
$\text{Order} = 1, \quad \text{Degree} = 1$ ✓ ½

Answer (i) $\frac{dA}{dt} = -kA$; **order 1, degree 1**

(ii) 1 mark

1 Separate and integrate.
$\frac{dA}{A} = -k\,dt \;\Rightarrow\; \log A = -kt + C$

2 At $t = 0$, $A = 400$, so $C = \log 400$. ✓ ½
$A = 400\,e^{-kt}$

3 At $t = 3$, $A = 300$.
$300 = 400\,e^{-3k} \;\Rightarrow\; e^{-3k} = \frac{3}{4} \;\Rightarrow\; e^{-k} = \left(\frac{3}{4}\right)^{1/3}$

4 Substitute e^{-k}. Hence shown. ✓ ½
$A = 400\left(e^{-k}\right)^t = 400\left(\frac{3}{4}\right)^{t/3}$

Answer (ii) $A = 400\left(\frac{3}{4}\right)^{t/3}$

(iii)(a) 2 marks

1 Put $t = 6$. ✓ ½
$A = 400\left(\frac{3}{4}\right)^{6/3} = 400\left(\frac{3}{4}\right)^2$

2 Simplify. ✓ ½
$A = 400 \times \frac{9}{16} = 225$

3 Amount removed $= 400 - 225 = 175$ mg. Find it as a percentage of the dose. ✓ ½
$\frac{175}{400} \times 100 = 43.75\%$

4 Hence 225 mg is present after 6 hours, and 43.75% of the dose has been removed. ✓ ½

Answer (iii)(a) **225 mg; 43.75% removed**

OR

(iii)(b) 2 marks

1 Put $A = 168.75$. ✓ ½
$168.75 = 400\left(\frac{3}{4}\right)^{t/3}$

2	Divide by 400. $\left(\frac{3}{4}\right)^{t/3} = \frac{168.75}{400} = \frac{27}{64} = \left(\frac{3}{4}\right)^{3}$	✓ ½
3	Compare powers. $\frac{t}{3} = 3 \Rightarrow t = 9$	✓ ½
4	Hence the next dose is given 9 hours after the first dose.	✓ ½

Answer (iii)(b) after 9 hours

Watch out: Writing $\frac{dA}{dt} = kA$ with $k > 0$; the medicine is being removed, so the rate is $-kA$.